

Source code:

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import heapq

class Patient:
    def __init__(self, patient_id, arrival_time, severity, doctor, arrival_no):
        self.patient_id = patient_id
        self.arrival_time = arrival_time
        self.severity = severity
        self.doctor = doctor
        self.arrival_no = arrival_no
        self.status = "Waiting"

    def __str__(self):
        return f"{self.patient_id} | Severity: {self.severity} | Arrival: {self.arrival_no} | Doctor: {self.doctor}"

class EmergencyPatientSystem:
    def __init__(self):
        self.heap = []
        self.patients = {}
        self.arrival_counter = 0

    def add_patient(self, patient_id, arrival_time, severity, doctor):
        if patient_id in self.patients:
            print("Patient already exists")
            return

        if severity < 1 or severity > 5:
            print("Invalid severity")
            return

        self.arrival_counter += 1

        patient = Patient(
            patient_id,
            arrival_time,
            severity,
            doctor,
            self.arrival_counter
        )

        self.patients[patient_id] = patient

        heapq.heappush(
            self.heap,
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        (severity, self.arrival_counter, patient_id)
    )

    print("Patient added successfully")

def next_patient(self):
    if not self.heap:
        print("No patients waiting")
        return

    _, _, patient_id = self.heap[0]
    print(self.patients[patient_id])

def treat_patient(self):
    if not self.heap:
        print("No patients waiting")
        return

    _, _, patient_id = heapq.heappop(self.heap)

    patient = self.patients[patient_id]
    patient.status = "Treated"

    print("Patient sent for treatment:")
    print(patient)

def search_patient(self, patient_id):
    if patient_id in self.patients:
        print(self.patients[patient_id])
    else:
        print("Patient not found")

def update_patient(self, patient_id, severity=None, doctor=None):
    if patient_id not in self.patients:
        print("Patient not found")
        return

    patient = self.patients[patient_id]

    if severity is not None:
        patient.severity = severity

    if doctor is not None:
        patient.doctor = doctor

    self.rebuild_heap()

    print("Patient updated successfully")

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def rebuild_heap(self):
    self.heap = []

    for patient in self.patients.values():
        if patient.status == "Waiting":
            heapq.heappush(
                self.heap,
                (
                    patient.severity,
                    patient.arrival_no,
                    patient.patient_id
                )
            )

def display_patients(self):
    if not self.heap:
        print("No patients waiting")
        return

    temp = sorted(self.heap)

    for _, _, patient_id in temp:
        print(self.patients[patient_id])

system = EmergencyPatientSystem()

system.add_patient("P001", "09:00", 3, "Dr. Arun")
system.add_patient("P002", "09:05", 1, "Dr. Priya")
system.add_patient("P003", "09:10", 2, "Dr. Kumar")
system.add_patient("P004", "09:15", 1, "Dr. Meena")

print("\nWaiting Patients:")
system.display_patients()

print("\nNext Patient:")
system.next_patient()

print("\nTreatment:")
system.treat_patient()

print("\nSearch:")
system.search_patient("P003")

print("\nUpdate:")
system.update_patient("P003", severity=1, doctor="Dr. Ravi")

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print("\nUpdated Queue:")  
system.display_patients()
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